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arXiv:2509.11928 (q-fin)

[Submitted on 15 Sep 2025 ([v1](#)), last revised 29 Oct 2025 (this version, v3)]

Meta-Learning Neural Process for Implied Volatility Surfaces with SABR-induced Priors

[Jirong Zhuang](#), [Xuan Wu](#)

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We treat implied volatility surface (IVS) reconstruction as a learning problem guided by two principles. First, we adopt a meta-learning view that trains across trading days to learn a procedure that maps sparse option quotes to a full IVS via conditional prediction, avoiding per-day calibration at test time. Second, we impose a structural prior via transfer learning: pre-train on SABR-generated dataset to encode geometric prior, then fine-tune on historical market dataset to align with empirical patterns. We implement both principles in a single attention-based Neural Process (Volatility Neural Process, VolNP) that produces a complete IVS from a sparse context set in one forward pass. On SPX options, the VolNP outperforms SABR, SSVI, and Gaussian process. Relative to an ablation trained only on market data, the SABR-induced prior reduces RMSE by about 40% and suppresses large errors, with pronounced

gains at long maturities where quotes are sparse. The resulting model is fast (single pass), stable (no daily recalibration), and practical for deployment at scale.

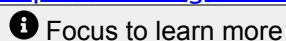
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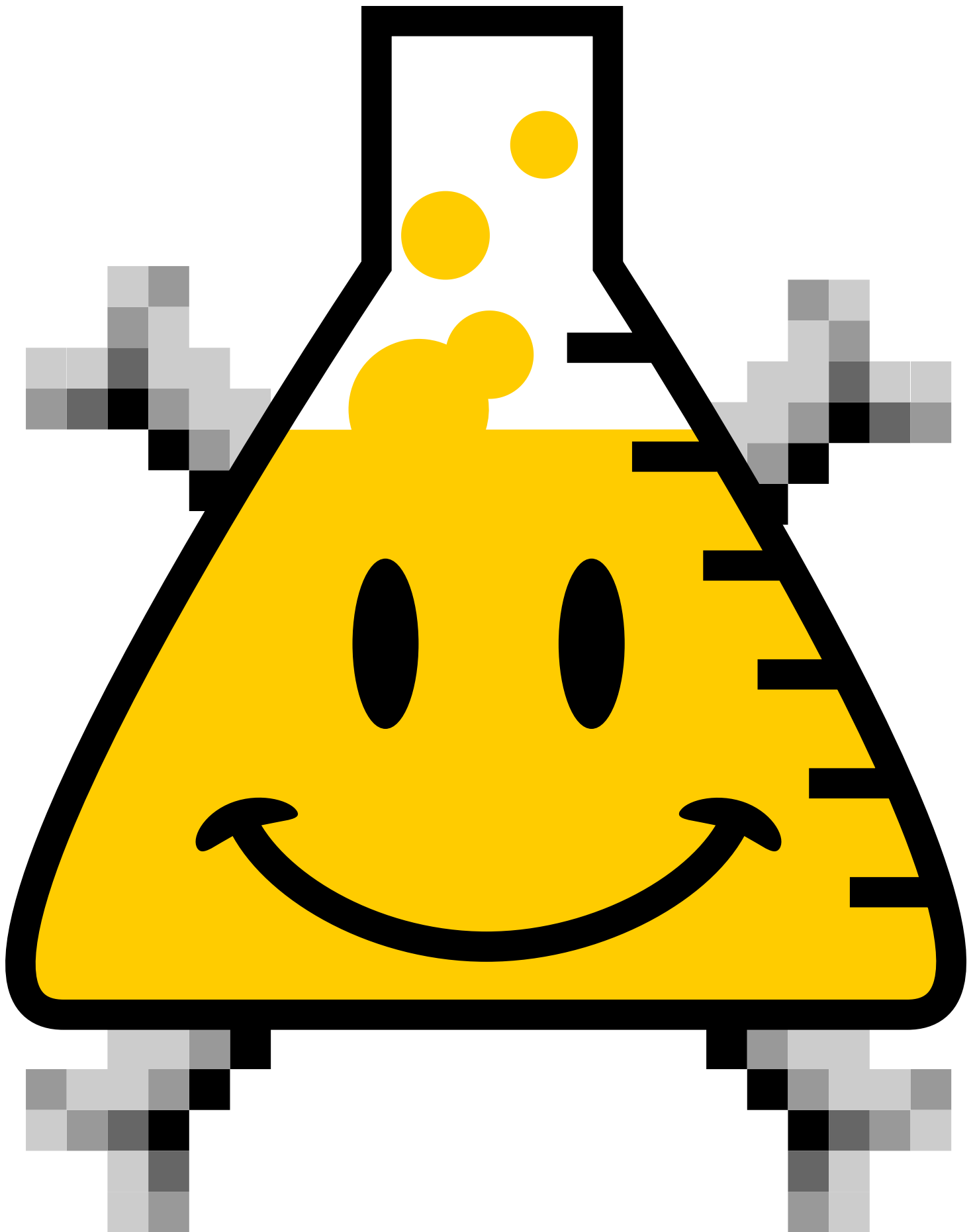
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